



DONATUS OMONDI

ELECTRICAL AND
ELECTRONICS ENGINEER

CONTACT INFO

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🌐 [Github](#)

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📍 Nairobi, Kenya

PROFILE

I'm an Embedded Systems Engineer with 2+ years of hands-on experience in firmware development, hardware-software integration, and PCB design for production and prototype systems. I have a strong background in STM32, ESP32 and RP2040 platforms, real-time systems using FreeRTOS, communication-heavy applications including CAN, Modbus, MQTT, and OCPP as well as experience in Zephyr. I also have slight experience in ASIC Design. I've demonstrated ability to identify and correct critical design flaws, implement scalable telemetry pipelines, and deliver robust embedded solutions in EV charging, IoT tracking, and smart agriculture domains.

EXPERIENCE

FEB 2025 - PRESENT

EMBEDDED HARDWARE ENGINEER

Geviton

Enterprises Ltd.

Full-time

- Designed and implemented BMS and charger drivers for STM32- and ESP32-based systems, enabling reliable EV battery charging and monitoring.
- Designed the PCB and oversaw production of charge controlling hardware
- Implemented OCPP-compliant firmware for EV charging infrastructure, supporting automated charging sessions and billing workflows.

EDUCATION

Sep 2019 – June 2024

Bachelor's Degree Electrical
and Electronics Engineering

University of Nairobi

Award: Second Class Honours

Upper Division

2015 – 2018

Maranda High School

Grade: A-

2007 – 2014

St. Hannahs Preparatory School

Grade: A

LANGUAGES

English



Swahili



REFEREE

✉ oscaror@uonbi.ac.ke

- Designed the PCB and oversaw production of tracker hardware
- Developed telemetry and data streaming pipelines over HTTP for remote monitoring and diagnostics.
- Identified and corrected critical PCB design flaws prior to manufacturing, preventing costly board respins and resolving long-standing hardware issues.
- Developed Python packages containing custom C extensions to meet performance and portability requirements across embedded and Linux-based systems.

JUL 2024 – JAN 2025

Geviton

Enterprises Ltd.

Full-time

GRADUATE TRAINEE

- Developed GSM and gas sensor drivers, enabling reliable data acquisition and communication for field-deployed devices.
- Created Python-based test and certification scripts, significantly improving efficiency during mass device certification.
- Designed and implemented firmware for vehicle and motorbike trackers with remote immobilization ("kill switch") functionality.
- Contributed firmware for remote land tracking devices used in asset and boundary monitoring.
- Reviewed and validated up to 6-layer PCB designs, issuing critical corrections before fabrication.

OCT 2023 – MAY 2024

STUDENT INTERN EMBEDDED FIRMWARE ENGINEER



Website

Geviton

Enterprises Ltd.

Part-time

- Developed GSM, RTC, and LoRa drivers for smart farming and battery metering hardware.
- Implemented secure telemetry pipelines (MQTTs) for cloud-based data aggregation.
- Reviewed PCB designs and provided firmware-informed feedback to improve future hardware revisions.
- Extended MicroPython firmware with low-level C/C++ modules, significantly reducing power consumption and improving runtime efficiency.

JUNE 2023 – OCT 2023

Geviton

Enterprises Ltd.

Full-time

STUDENT ON ATTACHMENT EMBEDDED SYSTEMS ENGINEER

Involved in development of hardware solutions. Responsibilities include pcb assembly in SMT line, assembly of final product in enclosures, firmware development for STM32, RP2040 and ESP8266 based boards

AUG 2022 – MAY 2023

Tafiti Nanosatellite

Part-time

STUDENT ON ATTACHMENT ELECTRICAL TEAM

Was part of the development of the ADCS Subsystem of the KSA-funded Tafiti 3U CubeSat

JUN 2022 – AUG 2022

Gearbox

Part-time

STUDENT ON ATTACHMENT

Created a flask server running on a raspberry pi on the local area network to handle user registration and image uploads. These images were used to train the AI module for face recognition access

CERTIFICATIONS

AUG 2022

HCIA Datacom

Huawei Certified ICT Associate Certificate

OTHER WORK

JAN 2026

W25Q_LFS

A lfs compliant helper for w25q chips

JUL 2024

RP2040 Deep
Sleep

Extended micropython source using C extensions to obtain deep sleep on the RP2040 chip

FEB 2023

Battery Charge
Control

Used Raspberry Pi Pico W to automate charging of a laptop through sub-net requests to flask server running in laptop to protect remote devices.

SEPT 2021 - AUG 2021

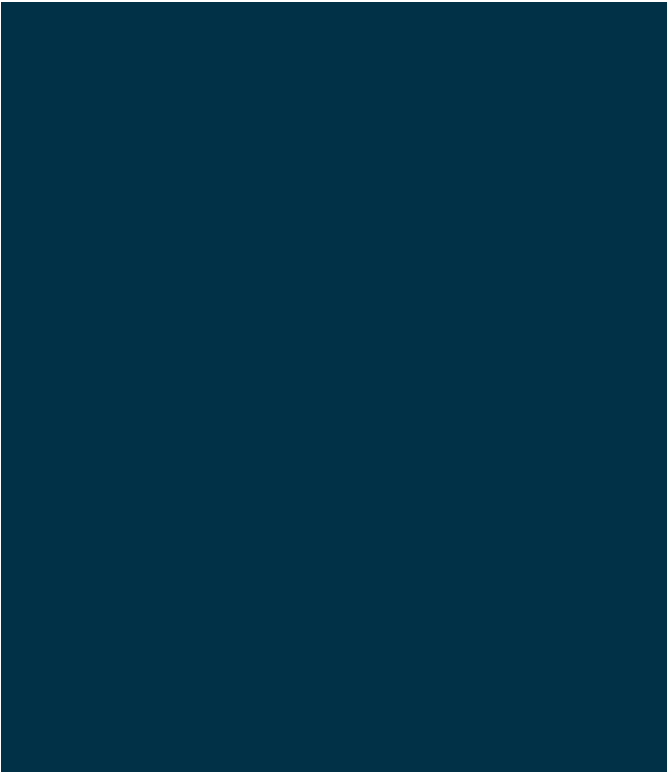
Fulfilment &
Expense Track

Created a web application to track deliveries, partial payments by customers to deliveries, expenses incurred by business and business intertransactions using Python, Flask and Javascript. This is currently being used by 3 businesses

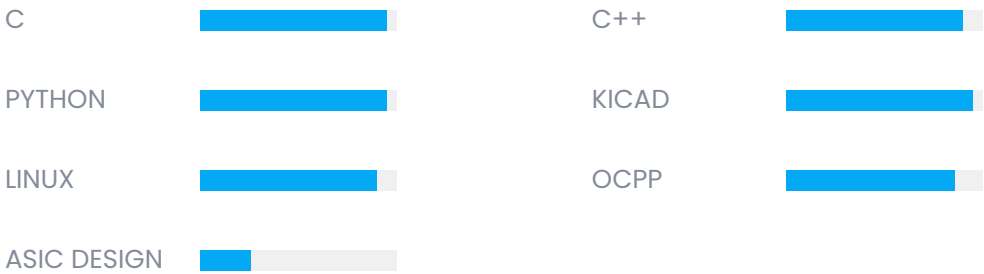
JULY 2021

Flask-dt python
package

Published a package to PYPI to dynamically display table records on user defined routes.



PROFESSIONAL SKILLS



INTERESTS

Programming Gaming Robotics